

Congenital Anomalies of the Kidney and Urinary Tract

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Congenital anomalies of the kidney and urinary tract (CAKUT) are a group of phenotypically diverse structural malformations characterized by defects in kidney and urinary tract development. CAKUT accounts for 20% to 30% of all developmental anomalies identified in the antenatal period and has a prevalence of 3 to 6 per 1000 births.¹ Although usually occurring in isolation, 30% are associated with extra-renal abnormalities, and CAKUT has been associated with more than 200 different syndromes, including Fraser syndrome, Kallman syndrome, and branchio-oto-renal (BOR) syndrome. CAKUT accounts for up to 50% of children with chronic kidney disease (CKD)² and is the leading cause of end-stage kidney disease (ESKD) in children, making up 34% to 43% of those requiring kidney replacement therapy (KRT).^{2,3} CAKUT also contributes significantly to the burden of kidney disease in young adults, accounting for 26% and 17% of individuals requiring KRT at age 18 to 21 years and 22 to 31 years, respectively,⁴ with a median age of KRT initiation of 31 years.⁵ Reduced nephron mass secondary to a solitary functioning kidney or posterior urethral valves (PUV) are associated with a worse kidney outcome, with concurrent vesico-ureteral reflux (VUR) a poor prognostic sign.^{6,7} Even with normal kidney function and blood pressure (BP) in adolescence, individuals with CAKUT have an increased lifetime risk of ESKD.⁸

CLINICAL PRINCIPLES

Congenital kidney and urinary tract anomalies may present in one of the following five settings:

1. Antenatal diagnosis by fetal ultrasound (US) screening
2. Failure to thrive in an infant or young child
3. Investigation of urinary tract infection (UTI)
4. An incidental finding in a child or adult
5. An adult with abnormal urinalysis, stones, hypertension, or reduced glomerular filtration rate (GFR)

The identification of these problems always poses the following questions:

- What is the cause?
- What is the natural history?
- Is surgical intervention required?

If there is a family history of structural malformations, a genetic cause may sometimes be identified, guiding surveillance for extra-renal manifestations, as in the case of *HNF1B*-related disease, as well as screening of first-degree relatives. Symptomatic UTIs are common in individuals with CAKUT and an increase in frequency of UTIs or deterioration in GFR should prompt investigations to exclude obstruction or bladder dysfunction that may require urologic intervention. Furthermore, as predicted by the Brenner hypothesis,⁹ small dysplastic kidneys with reduced GFR develop all the features of glomerular hyperfiltration. The onset of progressive kidney impairment signaled by increasing proteinuria and hypertension should

be treated according to standard CKD guidelines with initiation of renin-angiotensin system (RAS) blockade. The details of antenatal and pediatric management of these patients are beyond the scope of this chapter, which focuses on management in adolescence and adult life.

DEVELOPMENT OF THE KIDNEYS AND URINARY TRACT

The kidneys and urinary tract develop simultaneously from the endoderm-derived cloaca and intermediate mesoderm (Fig. 52.1).¹⁰ Kidney development can be divided into three phases: the pronephros, mesonephros, and metanephros. The pronephros develops 22 days after conception and forms a transient, rudimentary, and non-functioning system that degrades by day 28. It elongates caudally to meet the cloaca by day 26, becoming the mesonephric (Wolffian) duct, which ultimately contributes to the formation of the urinary bladder and male genital system (epididymis and caudal vas deferens).¹¹ The mesonephros is derived from the intermediate mesoderm and develops functioning tubules, which start to excrete urine, although most of these subsequently degenerate. By the fifth week of fetal life, the ureteral bud branches from the caudal part of the mesonephric duct into the metanephric mesenchyme to become the metanephros, the precursor to the adult kidney (see Fig. 52.1A). This process is mediated by glial cell line-derived neurotrophic factor (GDNF)/RET signaling, disruptions of which can result in varying phenotypes, including kidney agenesis. Reciprocal induction between the ureteral bud and the metanephric mesenchyme results in branching morphogenesis and elongation of the ureteral bud to form the collecting system and mesenchymal epithelial transformation of the metanephric mesenchyme to generate primitive nephrons (see Fig. 52.1B–D). The metanephros starts to function 6 to 10 weeks after fertilization, with nephrogenesis complete by 36 weeks. Sixty percent of nephrons are formed in the last trimester, which has important clinical implications for preterm and low-birth-weight infants, who have increased long-term risk for CKD.¹² The extent to which reduced nephron number contributes to this increased risk relative to exposure to nephrotoxic insults and acute kidney injury as a neonate is not yet understood.

The lower urinary tract is formed from the endodermal cloaca, which is divided by the urorectal septum into ventral and dorsal parts that develop into the urogenital sinus and rectum, respectively (see Fig. 52.1E). The urogenital sinus gives rise to the early bladder, the urethra and vestibule of the vagina in females, and the posterior urethra in males. Growth of the anterior abdominal wall between the allantois and the urogenital membrane is accompanied by an increase in size and capacity of this bladder precursor. The allantois remains attached to the apex of the fetal bladder and extends into the umbilical root, although it loses its patency and persists as the urachal remnant, the

Development of the Urinary Tract

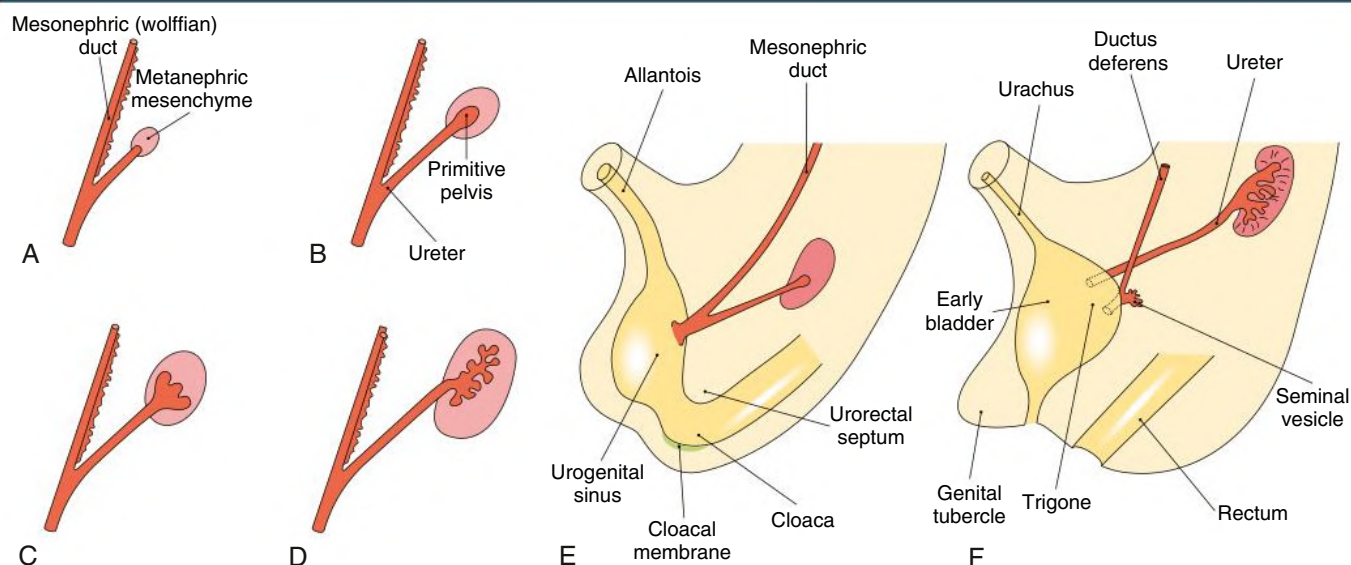


Fig. 52.1 Development of the Urinary Tract. Growth and development of the ureter, pelvis, and calyces are shown in parts A to D. (A) The metanephric kidneys first become detectable as small areas in the mesoderm close to the aorta. The primitive epithelial ureter buds off from the mesonephric duct and makes contact with the metanephric mesenchyme. (B) Under the influence of signals from the ureter, the mesenchyme condenses and proliferates around the ureteral tip, with simultaneous elongation and branching of the ureteral tip. (C–D) A primitive pelvis appears, then branches to form the divisions of the calyces. The branching process continues, with the epithelial system eventually differentiating into the nephrons of the kidney parenchyma. As the fetus grows, the kidney ascends because of the continuous rostral growth. (E) Growth and development of the cloaca during weeks 5 to 6 of gestation. (F) Growth and development of the urogenital sinus into bladder and outflow tract during weeks 8 to 9.

median umbilical ligament, which connects the bladder to the umbilicus (see Fig. 52.1F). By the seventh week, there is a separate opening of the distal mesonephric duct into the bladder at what will become the vesico-ureteral opening and the area known as the *trigone*. At the same time the paramesonephric (müllerian) ducts start to regress in males and fuse in females to become the uterovaginal cord, which opens into the urogenital sinus and will go on to develop into the vagina.

PATHOGENESIS

Familial clustering, monogenic syndromes associated with urinary tract malformations, and animal models suggest a strong genetic basis for CAKUT. US screening of asymptomatic first-degree relatives reveals structural urinary tract malformations in up to 23%.¹³ The increasing availability of next-generation sequencing has led to the identification of over 50 monogenic CAKUT-causing genes, but genotype-phenotype heterogeneity and incomplete penetrance mean that a single-gene cause has been identified in less than 20% of individuals.¹⁴ In addition, rare copy-number variation (CNV) has been reported in up to 40% of individuals with CAKUT, suggesting that gene dosage may also be an important mechanism.¹⁵ Environmental factors, including preexisting and gestational maternal diabetes¹⁶ and increases or decreases in maternal vitamin A levels,¹⁷ have also been associated with an increased risk of anomalies, and it is likely that CAKUT results from the complex interplay of genetics and maternal and environmental factors.

Kidney and urinary tract development are tightly regulated by a network of transcription factors, growth factors, and adhesion molecules. Pathogenic variants in genes encoding all classes of these molecules,

affecting different stages of nephrogenesis, have been identified in patients with CAKUT (Table 52.1). Variants in the transcription factors *HNF1B* (hepatocyte nuclear factor 1B) and *PAX2* (paired box gene 2) are two of the most common causes of CAKUT (both syndromic and isolated) and are associated with cystic kidneys and kidney hypodysplasia (Table 52.2), respectively.

- *HNF1B* mediates the development of the kidneys, liver, pancreas, and urinary tract. Haploinsufficiency resulting from whole-gene deletions (typically a 1.4 Mb 17q12 deletion) or pathogenic heterozygous variants can cause kidney cysts and diabetes syndrome (KCAD). A high frequency of de novo deletions means there may not be a family history. *HNF1B*-related disease is associated with a wide range of CAKUT phenotypes: kidney hypodysplasia, cystic kidneys, single and horseshoe kidneys, malformations of the collecting system, and autosomal dominant tubulointerstitial kidney disease (ADTKD-HNF1B).¹⁸ Extrarenal manifestations include mature-onset diabetes of the young [MODY] type 5, pancreatic hypoplasia, genital malformations, hyperuricemia and early gout, abnormal liver function, and hypomagnesemia, highlighting the utility of a molecular diagnosis in guiding evaluation for possible nonrenal abnormalities.
- *PAX2* is a key regulator of multiple steps of kidney development, including ureteric bud induction, branching morphogenesis, and nephron differentiation.¹⁹ Heterozygous variants in *PAX2* were first discovered in patients with renal-coloboma syndrome presenting with kidney hypodysplasia, optic nerve abnormalities, and hearing loss, but variants have now been identified in a wide range of phenotypes, including adult-onset focal segmental glomerulosclerosis (FSGS).²⁰

TABLE 52.1 Monogenic Causes of CAKUT

Inheritance	Gene	Associated Phenotype
Autosomal recessive	<i>CHRM3</i>	Prune-belly like syndrome
	<i>FRAS1</i>	Fraser syndrome, kidney agenesis
	<i>FREM1</i>	Bifid nose, kidney agenesis
	<i>FREM2</i>	Fraser syndrome, kidney agenesis
	<i>GFRA1</i>	Bilateral kidney agenesis
	<i>GRIP1</i>	Fraser syndrome, kidney agenesis
	<i>HPSE2</i>	Urofacial syndrome, neurogenic bladder
	<i>ITGA8</i>	Bilateral kidney agenesis
	<i>LRIG2</i>	Urofacial syndrome, neurogenic bladder
	<i>TRAP1</i>	VACTERL
Autosomal dominant	<i>BNC2</i>	Congenital lower urinary tract obstruction/PUV
	<i>EYA1</i>	Branchio-oto-renal syndrome
	<i>GATA3</i>	HDR syndrome
	<i>GLI3</i>	Pallister-Hall syndrome
	<i>GREB1L</i>	Kidney agenesis
	<i>HNF1B</i>	Kidney cysts and diabetes syndrome
	<i>PAX2</i>	Renal-coloboma syndrome, FSGS
	<i>PBX1</i>	Kidney hypodysplasia
	<i>RET</i>	Kidney agenesis
	<i>SALL1</i>	Townes-Brocks syndrome, kidney hypodysplasia
	<i>SIX5</i>	Branchio-oto-renal syndrome
	<i>TBX18</i>	PUJO
	<i>ZMYM2</i>	Syndromic CAKUT
X-linked recessive	<i>ANOS1</i>	Kallmann syndrome, kidney agenesis

CAKUT, Congenital anomalies of the kidney and urinary tract; FSGS, focal segmental glomerulosclerosis; HDR, hypoparathyroidism, sensorineural deafness, and kidney disease; PUJO, pelvi-ureteral junction obstruction; PUV, posterior urethral valves; VACTERL, vertebral defects, anal atresia, cardiac defects, tracheo-esophageal fistula, renal (kidney) anomalies, and limb anomalies.

KIDNEY PARENCHYMAL MALFORMATIONS

Congenitally abnormal kidneys may be enlarged or small, cystic or dysplastic, and absent or misplaced. These conditions were traditionally discussed based on findings on intravenous urography (IVU) but are now primarily investigated with US, computed tomography (CT), and magnetic resonance imaging (MRI). Functional assessment of obstruction and VUR are carried out using ^{99m}Tc -mercaptoacetyltriglycine (MAG3) scintigraphy and voiding cystourethrogram (VCUG), respectively.

Kidney Dysplasia

Table 52.2 presents the range of dysplastic and other malformations of the kidney. Abnormalities of the ureter, bladder, and urethra are often associated with kidney dysplasia, and evidence of concurrent VUR should always be sought. All types of kidney dysplasia can also occur as isolated developmental anomalies and although typically producing small, irregular kidneys, they can also be cystic or multicystic. US features of kidney dysplasia include increased echogenicity, poor corticomedullary differentiation, and parenchymal cysts.

TABLE 52.2 Definitions of Kidney Malformations

Term	Characteristics
Kidney agenesis	Absence of the kidney or an identifiable metanephric structure.
Kidney aplasia	Severe dysplasia with extremely small kidney, sometimes identifiable only by histologic examination.
Kidney dysplasia	Abnormal differentiation of kidney parenchyma with development of abnormal structures, including primitive ducts surrounded by collars of connective tissue, metaplastic cartilage, variety of nonspecific malformations such as preglomeruli of fetal type, and reduced branching of collecting ducts with cystic dilations and primitive tubules. Dysplastic kidneys often contain cysts.
Kidney hypoplasia	Significantly reduced kidney mass with either normal or reduced (oligomeganephronia) nephron number without evidence of maldevelopment of parenchyma.
Kidney hypodysplasia	Reduced kidney mass and nephron number with dysplastic features. Previously thought to be secondary to scarring from reflux or reflux nephropathy but now considered to be primary dysplasia with associated reflux.
Kidney multicystic dysplasia	Severe cystic dysplasia with extremely enlarged kidney full of cystic structures; occurs as an isolated kidney lesion in response to ureteral atresia and urethral obstruction; 10% of patients have a family history.

Kidney Hypoplasia

Kidney hypoplasia is defined as a congenitally small kidney (two standard deviations below the expected mean) that lacks evidence of either parenchymal maldifferentiation (kidney dysplasia) or acquired disease sufficient to explain the reduced size. The term is often used loosely and includes small kidneys with a normal number of nephrons as well as oligomeganephronia. This is a type of kidney hypoplasia resulting from a congenital reduction in the number of nephrons. It results from arrested development of the metanephric blastema at 14 to 20 weeks of gestation with subsequent hypertrophy of glomeruli and tubules in the kidney. The hypertrophy and hyperfiltration result in progressive nephron injury and sclerosis later in life. Oligomeganephronia is recognized on kidney biopsy by the large size of the glomeruli and tubules and the small number of glomeruli seen despite a good core of kidney cortex.

Differential Diagnosis of Scarred Kidneys

Kidney hypodysplasia versus reflux. Progressive scarring and kidney failure were once considered to be caused by chronic parenchymal infection (so-called “chronic pyelonephritis”) and were regarded as a consequence of VUR. The 1980s, however, saw a retreat from the paradigm of the primary role of infection, and emphasis was placed on scarring as a result of reflux and the progressive nature of the glomerular lesion associated with glomerular hypertension (or hyperfiltration), so-called “reflux nephropathy” (see Chapter 62). Current thinking is that scarring often follows from kidney dysplasia, and that the reflux is a secondary or incidental feature (Fig. 52.2). Thus, irregular kidneys with normal-caliber ureters are more likely to be caused by primary dysplasia, and no evidence of VUR may be seen.

Kidney scarring in adults. A practical clinical problem is the differential diagnosis of scarred, asymmetric kidneys. Genetic testing

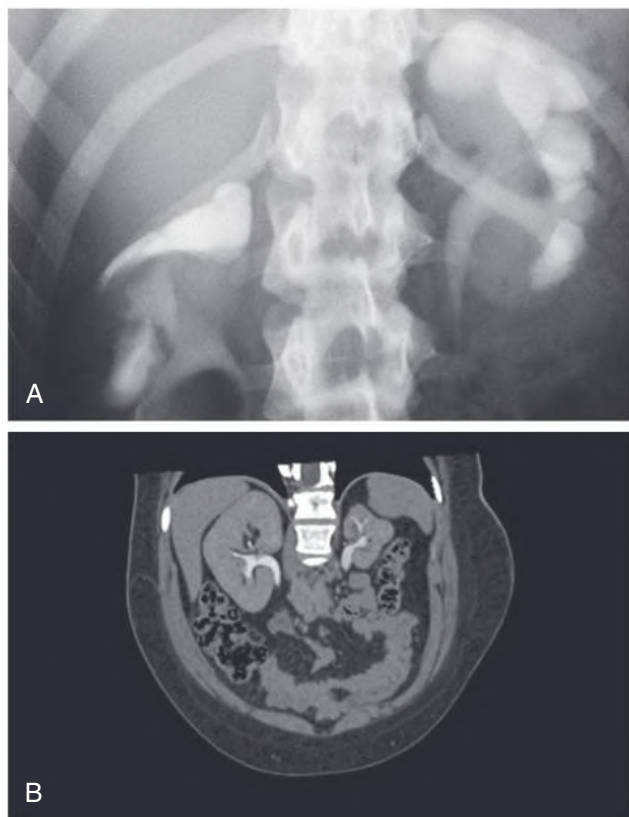


Fig. 52.2 Kidney Dysplasia. (A) Intravenous urogram (IVU) shows gross bilateral scarring in a 20-year-old woman who has been followed since the age of 2 years. Progressive scarring has been observed in the absence of urinary tract infections and obstruction. This probably represents primary kidney dysplasia. (B) Computed tomography IVU shows gross scarring of right kidney. (Courtesy Dr. A. Kirkham, University College Hospitals, London.)

may be useful in younger patients, particularly in those with a family history of kidney disease. With older patients, the differential diagnosis of scarred or “lumpy, bumpy” kidneys widens. Sometimes attributed to other diagnoses, including analgesic nephropathy, this appearance is often designated reflux nephropathy. In older patients, multiple scarring from atheromatous arterial disease and embolization of the kidney is an increasingly important cause of kidney failure. The diagnosis has historically been made by the radiologic features on IVU, but CT and magnetic resonance (MR) urography are now the gold standard, with scarring best demonstrated by technetium 99m -labeled dimercaptosuccinic acid (99m Tc-DMSA) scintigraphy.

Absent Kidneys

Unilateral Kidney Agenesis

Complete absence of one kidney occurs in approximately 1 in 3000 live births and may result from involution of a multicystic dysplastic kidney (MCDK) or true kidney agenesis. It can be inherited as an autosomal dominant trait with incomplete penetrance and variable expression, associated with heterozygous variants in *RET* and *GREB1L*, or biallelic variants in *FRAS1* and *FREM2*.¹

Agenesis occurs when the ureteric bud fails to develop and induce formation of the metanephros; typically, there is no ureter, and the ipsilateral half of the bladder trigone is missing. A blind-ending ureter may be seen in cases of MCDK. The remaining kidney is usually hypertrophic, but it may be ectopic, malrotated, or hydronephrotic with a megaureter and associated VUR. The more severe the dysplasia of the

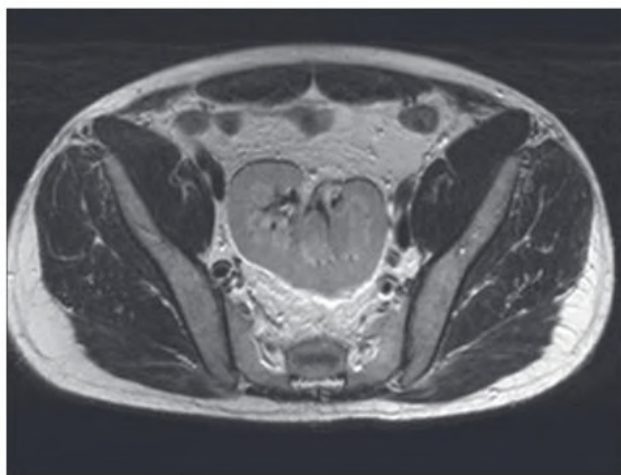


Fig. 52.3 Single Pelvic Kidney. Magnetic resonance scan, transverse section, shows single midline pelvic kidney. (Courtesy Dr. A. Kirkham, University College Hospitals, London.)

remaining kidney is, the earlier the presentation. The ipsilateral testis and seminal tract are usually absent, and in 10% of cases, the adrenal gland is also missing. Females can have an absent fallopian tube or ovary or malformation of the vagina or uterus. Other associations seen in 30% to 50% of those affected include imperforate anus and malformations of the vertebrae and cardiovascular system.

Normality of the single kidney should be confirmed by 99m Tc-DMSA scintigraphy and isotopic GFR. Compensatory glomerular hyperfiltration means that lifelong follow-up is necessary to monitor for proteinuria, hypertension, and CKD, which can be seen in up to 20% of individuals. Evaluation for urogenital anomalies is recommended in all first-degree relatives of individuals with kidney agenesis.

Bilateral Kidney Agenesis

Bilateral kidney agenesis is lethal. Although usually sporadic, biallelic variants in *ITGA8* and *GFRA1* have been reported. It is associated with pulmonary hypoplasia and a characteristic facial appearance (Potter facies) caused by intrauterine compression, which is a consequence of oligohydramnios. The prevalence is about 1 in 10,000 births, with risk for occurrence in siblings of about 3%, unless there is a family history of agenesis, in which case risk rises to 15% to 20%.

Misplaced Kidneys

Kidney Ectopia, Malrotation, and Crossed Fused Kidneys

The starting position of the fetal kidney is deep in the pelvis. Kidneys that fail to ascend properly and therefore remain lower than usual occur in 1 in 1000 births (Fig. 52.3). By the eighth week of gestation, the kidney ascends, and the pelvis comes to face more medially. The most common anomaly is for the pelvis to face forward. Those that do not migrate past the pelvic brim are called pelvic kidneys. The more ectopic the kidney is, the more severe the rotation is and the more abnormal the appearance. In more than 90% of ectopia, there is fusion of both kidneys, which may cross the midline (crossed fused ectopy). This is best visualized on CT or MR urography (Fig. 52.4). Symptoms and complications, if any, are usually caused by associated VUR, but in the absence of additional anomalies or obstruction, no follow-up is required.

Horseshoe Kidney

If both kidneys are low, they may fuse at the lower pole and are usually drained by two ureters (Fig. 52.5). The kidneys lie lower than normal,

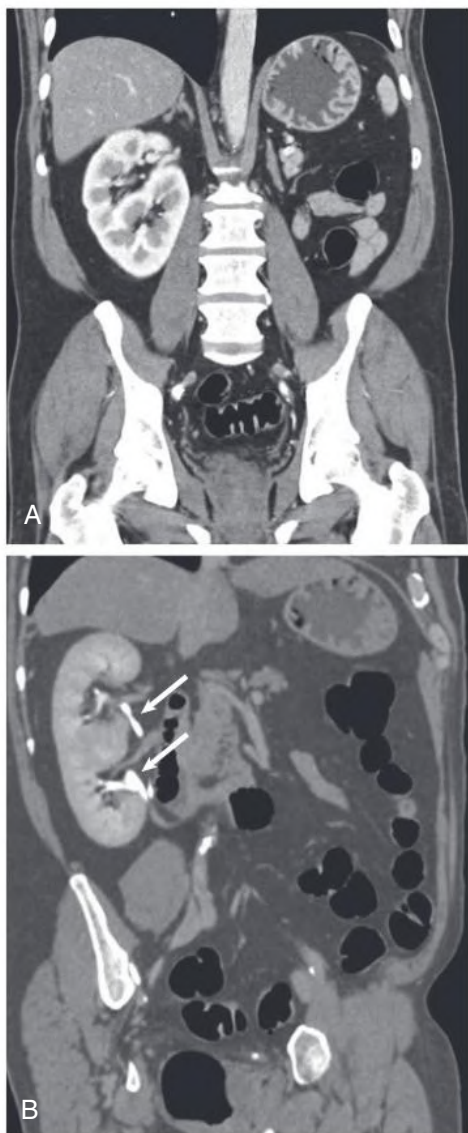


Fig. 52.4 Crossed Fused Ectopia. (A) Magnetic resonance scan shows fused kidneys on right. (B) There are two ureters (arrows). (Courtesy Dr. A. Kirkham, University College Hospitals, London.)

and further ascent is prevented by the root of the inferior mesenteric artery. Horseshoe kidney occurs in 1 in 600 to 1800 births and is more common in males (2:1). VUR and pelvi-ureteral junction (PUJ) obstruction are commonly seen with kidney calculi also occurring in up to 20%. Hydronephrosis in a horseshoe kidney should be investigated with a ^{99m}Tc -MAG3 renogram to differentiate obstruction from urinary stasis and urologic opinion sought if necessary. In most cases, individuals remain asymptomatic and do not require nephrologic follow-up.

Calyceal Abnormalities

Hydrocalyx and Hydrocalycosis

Dilated calyces are usually caused by obstruction. Focal dilation also can be caused by congenital infundibular stenosis, extrinsic compression from vessel or tumor, stones, or tuberculosis. If obstruction is excluded, the appearance is likely to be a congenital abnormality and can be an incidental finding. Moreover, if the GFR is normal and the divided function of the kidneys is 50:50, surgery to improve the anatomy should not be attempted.

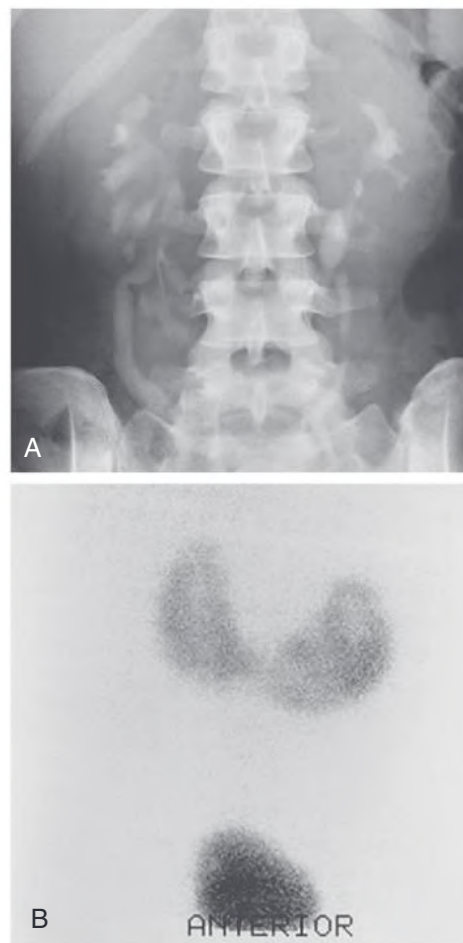


Fig. 52.5 Horseshoe Kidney. (A) Intravenous urogram soon after pregnancy in a 25-year-old woman shows the horseshoe kidney joining in the midline as well as dilated ureters as a transient effect of pregnancy. (B) Dimercaptosuccinate scan shows a horseshoe kidney and the urinary bladder.

Megacalycosis

In megacalycosis, there is bizarre dysplasia of the calyceal system with an increase in the number of calyces. There is no obstruction, and the cause is malformation of kidney papillae. Megacalycosis is congenital, usually unilateral, and an incidental finding. It is much more common in males (6:1). Bilateral disease is confined to males and segmental, unilateral disease to females, which suggests an X-linked partially recessive gene with reduced penetrance in females. There may be an associated ipsilateral segmental megaureter, usually affecting the distal third.

Calyceal Diverticulum (Calyceal Cyst)

A calyceal diverticulum is a cavity peripheral to a minor calyx that is not a closed cyst but rather is connected to the calyx by a narrow channel (Fig. 52.6). It is usually an incidental finding and may manifest with symptoms relating to stones or infection within the cavity.

Bardet-Biedl Syndrome

Multiple calyceal clubbing and calyceal diverticula are the characteristic features of the kidney dysplasia seen in Bardet-Biedl syndrome (formerly known as *Laurence-Moon-Biedl syndrome*). This rare autosomal recessive condition is characterized by retinitis pigmentosa, polydactyly, obesity, hypogonadism, learning difficulties, and kidney anomalies in 80%. Calyceal malformation is associated with parenchymal

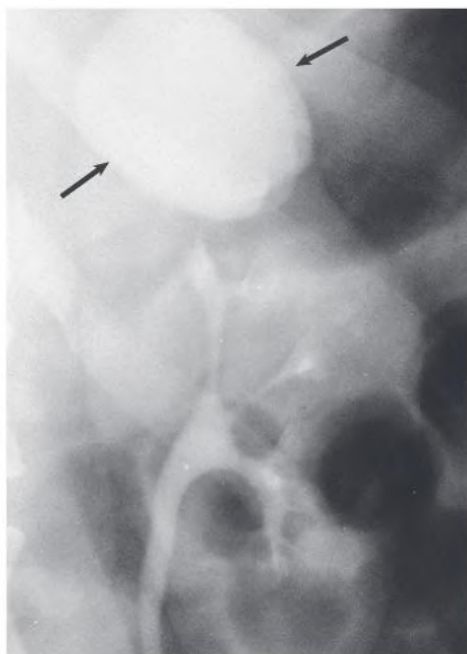


Fig. 52.6 Calyceal Cyst. Intravenous urogram shows an upper pole calyceal cyst filled with contrast (arrows). Plain abdominal radiograph showed a group of stones in the floor of the cyst.

dysplasia, fetal lobulation, kidney cysts, horseshoe kidneys, and VUR; kidney failure in early adult life is common. Bardet-Biedl syndrome is caused by a defect of the basal body of ciliated cells,²¹ and mutations in 22 genes coding for different proteins located in the basal body and primary cilia of the cell have been reported, making the syndrome an archetypal ciliopathy.

COLLECTING SYSTEM ABNORMALITIES

Pelvi-ureteral Junction Obstruction

PUJ obstruction occurs in 1 in 1500 live births and is one of the most common causes of antenatally detected hydronephrosis. It is seen more frequently in males (2:1), preferentially affecting the left kidney. The condition is usually congenital, caused by intrinsic stenosis or kinking of the proximal ureter at the PUJ, but can have an acquired mechanical basis as a result of external compression from an aberrant or accessory lower pole vessel or adhesions. Associated abnormalities are common, and up to 50% of infants have another urologic abnormality, such as contralateral PUJ obstruction, contralateral dysplastic and multicystic kidney, minor degrees of VUR, and contralateral kidney agenesis.

Older children can present with an abdominal mass or with intermittent flank pain that worsens during a brisk diuresis (e.g., after consumption of caffeine or alcohol), a phenomenon called Dietl's crisis. Hematuria secondary to mild trauma and UTIs may also occur. Hypertension is unusual but can occur temporarily after surgical correction.

^{99m}Tc-MAG3 diuretic renogram should be used to differentiate between significant obstruction that requires surgical correction and congenital ectasia of the renal pelvis, in which case surgery is not indicated (Fig. 52.7). Kidneys with good function can generally be left alone, and pyeloplasty is indicated only when function is clearly deteriorating (>10% decrease in split kidney function) or for symptomatic relief of pain, infection, or stones.²²

Algorithm to Exclude Obstruction

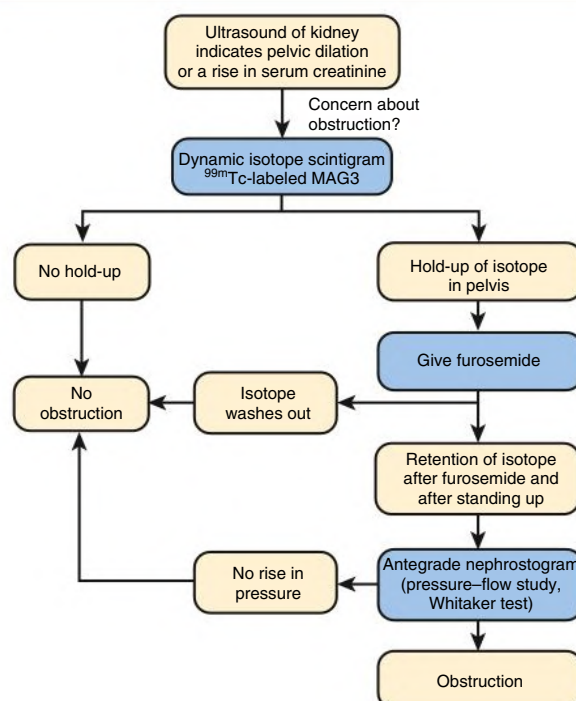


Fig. 52.7 Algorithm to Exclude Pelvi-Ureteral Junction (PUJ) Obstruction. Diagnostic path used to differentiate between significant PUJ obstruction and congenital ectasia of renal pelvis. MAG3, Mercaptoacetyltriglycine.

Duplex Collecting System

Partial or complete duplication of the ureter and renal pelvis is a common anomaly, with an incidence of about 1 in 150 births; unilateral duplication is six times more frequent than bilateral. It is more common in females. If duplication has been detected in a patient, the likelihood of another sibling with duplication rises to 1 in 8.

Pathogenesis

If the ureteral bud bifurcates after its origin from the mesonephric duct but arises at a normal site, an incomplete ureteral duplication with a Y ureter will develop. Complete ureteral duplication occurs if there are two ureteral buds, one in the normal location and the other in a low position. The normal bud ends in a correct site on the trigone in the bladder and is nonrefluxing. The lower bud, representing the ureter of the lower pole of the kidney, ends in the bladder as a lateral orifice with a short submucosal tunnel. The lower pole ureter is therefore often associated with VUR.

If there are two ureteral buds, one with a normal location and one with a high position, the upper ureter can be incorporated into the developing bladder, ending more distally and medial to the normal one, or inserted ectopically elsewhere. The ectopic ureter can result in obstruction or VUR, and there is often associated dysplasia of the upper pole moiety.

Clinical Manifestations

In most adult patients, ureteral duplication is asymptomatic and causes no long-term problems. Children with ureteral duplication often have VUR. The spontaneous disappearance of reflux is less common in duplex ureters than in patients with a single ureter. Duplex ureters are

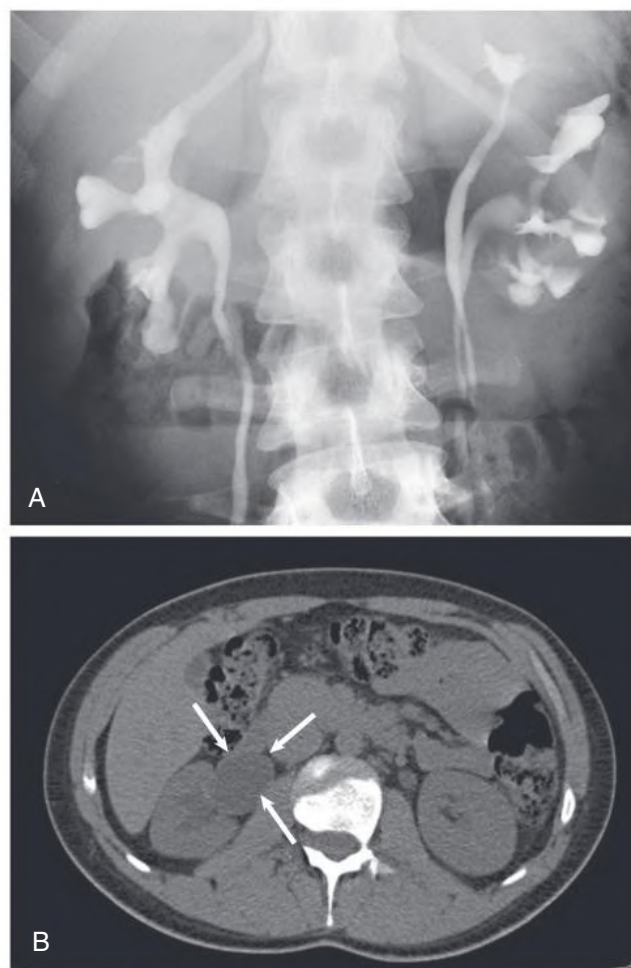


Fig. 52.8 Duplex Kidney. (A) Intravenous urogram shows a duplex left kidney. The lower pole is scarred and shows evidence of reflux damage. The two ureters enter the bladder separately, with the lower pole ureter in the abnormal location. The right kidney also shows features of reflux, with clubbing of the calyces and some scarring. (B) Computed tomography scan shows an isolated right-sided megaureter (arrows).

best diagnosed by CT urogram. PUJ obstruction of the ureter draining the lower pole of the kidney can occur. Associated conditions, such as ectopic ureters and ureterocele (see later discussion), usually cause problems in early life and therefore have been addressed by adolescence. Upper pole scarring is associated with an ectopic ureter and lower pole scarring with VUR (Fig. 52.8A).

Ectopic Ureters

Ectopic ureters are almost always associated with ureteral duplication and insert along the developing mesonephric system; 10% are bilateral and there is a female-to-male ratio of 7:1. The ectopic ureter comes from the upper pole and inserts into the bladder more distally and toward the bladder neck or opens into the upper urethra. In females, the ureter may end in the bladder neck or upper urethra, vagina, cervix, or uterus, and patients present with incontinence, UTIs, or a persistent vaginal discharge, particularly if the external sphincter is damaged, as during labor.

Ectopic ureters are rare in males and manifest as UTI. Usually, there is a single ureter associated with a dysplastic kidney, which ends in the posterior urethra, seminal vesicle, or vas deferens. Males are usually continent because the ureter is proximal to the external sphincter.

Ectopic ureters are best visualized by CT or MR urography. A voiding cystourethrogram shows reflux into the lower pole of the kidney in 50% of patients.

Ureterocele

Ureteroceles are cystic dilations of the terminal segments of the ureter. Ureteroceles affect females more than males (4:1), and 10% are bilateral. Ectopic ureters and ureters with ureteroceles frequently (80%) drain the upper pole of a duplex collecting system and are often associated with dysplastic or nonfunctional kidney tissue. These usually present in childhood with infection; when large, they can obstruct the bladder neck or even the contralateral ureter. In adults, ureteroceles typically manifest with stones in the lower ureter.

First-line management of ureteroceles is endoscopic decompression, although partial nephroureterectomy, ureteral reimplantation, or ureterocele excision and bladder neck reconstruction may be required. There are usually no medical sequelae.

Megaureter

Primary megaureter occurs in 3 to 4 per 10,000 live births, is seen more frequently in males, and is bilateral in 30% to 40% of cases. Isolated dilation of the ureter does not necessarily imply obstruction. There are three broad groups of conditions with widely dilated ureters, as follows:

1. *Obstruction of the ureter itself.* This may be intrinsic (e.g., stone) or extrinsic (e.g., retroperitoneal fibrosis); it is not associated with reflux.
2. *Bladder outflow obstruction, with secondary ureteral obstruction.* Examples include a neuropathic bladder and posterior urethral valves; this may or may not be associated with reflux.
3. *A dilated but nonobstructed ureter.* This often occurs without reflux, and there can be normal kidney function; this may be caused by an adynamic segment of the lower ureter (see Fig. 52.8B).

Pathogenesis

In the normal ureter, there is a characteristic helical orientation of muscle fibers. When the megaureter is secondary to bladder outflow obstruction, there is muscle hyperplasia and hypertrophy of the ureteral wall. In primary megaureter, a variety of abnormalities of muscle orientation are described or there may be absence of muscle fibers at the proximal end of the undilated segment. Electron microscopy shows an increase in collagen between the muscle bundles at the level of the obstructing segment. Obstruction appears to be caused by a failure of peristalsis through the distal ureteral segment.

Clinical Manifestations

Most cases of megaureter associated with obstruction present in childhood with severe infections, often complicated by septicemia. These patients have a high incidence of other congenital abnormalities. In less severe cases or with no obstruction, patients can present with abdominal pain, loin pain, hematuria, and UTI. Kidney stones can form easily in the dilated systems. The exclusion of obstruction is often established only by an antegrade pressure-flow study (Whitaker test), in which a nephrostomy is placed in the renal pelvis and contrast material infused at 10 mL/min.²³

Treatment

It must be definitively determined whether obstruction exists (see Fig. 52.7). The current view is that patients with asymptomatic nonobstructed disease should be managed conservatively, and most do well with this approach, with spontaneous remission rates of up to 85%.

Vesico-ureteral Reflux

VUR describes the retrograde passage of urine from the bladder to the upper tract. Such patients fall into two broad groups. First, there is a group of patients who appear to have normal bladders without outflow obstruction and normal caliber ureters when not micturating, described as having primary VUR resulting from a congenitally shortened intravesical ureter. The remainder have some form of anatomic or functional bladder outflow dysfunction that causes secondary VUR and dilated upper urinary tracts, the most common cause of which is PUV in males. VUR has an incidence of around 1%, is seen more frequently in females, and usually spontaneously regresses with age. Although previously believed to cause “reflux nephropathy” or “chronic pyelonephritis,” where recurrent infection leads to acquired kidney scarring, advances in genetics and developmental biology indicate that kidney impairment is usually the result of associated primary kidney dysplasia rather than reflux itself.

Pathogenesis

There is a strong familial association with primary VUR; 27% of siblings and 36% of offspring of affected parents have been found to have VUR.²³ Linkage and genome-wide studies have yet to identify any robust genetic associations, highlighting the heterogeneity of this often-undiagnosed condition.

Clinical Manifestations

VUR is usually diagnosed after a febrile UTI by voiding cystourethrogram (VCUG). Severity is graded I to V with grades IV and V (high-grade VUR) defined by gross dilation of the ureter and collecting system, blunting of the calyces, and tortuosity of the ureter (see Chapter 62). The higher the grade, the less likely it is that spontaneous resolution will occur and the higher the prevalence of recurrent symptomatic UTI. US and DMSA imaging should be undertaken to look for associated kidney anomalies, differential function, and cortical defects. Concurrent lower urinary tract symptoms may also occur.

Treatment

Management of VUR aims to prevent recurrent infections and preserve kidney function. In children, antibiotic prophylaxis reduces the risk of recurrent UTI but has not been shown to have an effect on kidney scarring.²⁴ Indications for surgical intervention include grade III to V reflux, which does not spontaneously remit and is associated with breakthrough infections on prophylaxis. Endoscopic injection of a bulking agent around the intravesical ureter (DEFLUX) or open ureteric reimplantation are valid options, although VUR is more likely to recur after endoscopic correction. Current guidelines do not recommend the screening of siblings or offspring because of uncertainty regarding the clinical importance of asymptomatic VUR.

BLADDER AND OUTFLOW DISORDERS

Prune-Belly Syndrome

Prune-belly syndrome (PBS) nearly always affects males and is characterized by the absence of anterior abdominal wall muscles, bizarre malformations of the urinary tract with gross dilation of the bladder and ureters, and bilateral undescended testes.²⁵ When the disorder is diagnosed early, kidney outcome is related to the degree of kidney dysplasia. There are incomplete forms of PBS (pseudoprune). Rarely, a similar megacystis or megaureter may be seen in a male or female patient.



Fig. 52.9 Prune-Belly Syndrome. Note the lax abdominal musculature leading to a pot-bellied appearance. There is also marked thoracic cage deformity. (Courtesy Prof. C.R.J. Woodhouse, University College Hospital, London.)

Pathogenesis

The incidence of PBS varies from 1 in 35,000 to 1 in 50,000 live births. Some familial cases have been reported, implicating genes affecting smooth muscle function. Biallelic loss-of-function mutations in *CHRM3*, a muscarinic acetylcholine receptor involved with parasympathetic-mediated detrusor contraction, have been identified in one family with prune-belly-like syndrome.²⁶ There is also evidence for a primary, localized arrest of mesenchymal development, supported by the lack of prostatic differentiation; the epithelial element in the prostate is absent or hypoplastic. Ultrastructure studies of the ureter show massive replacement of smooth muscle with fibrous and collagen tissue and the absence of nerve plexuses. An almost identical syndrome can result from fetal urethral obstruction, including urethral atresia.

Clinical Manifestations

The prognosis depends on the degree of kidney dysplasia and injury. Three prune-belly groups can be distinguished.²⁵ In group I (20%), complete urethral obstruction causes stillbirth or neonatal death. In group II (20%), acute, early presentation requires diversion and reconstruction. In group III (60%), good health and kidney function continue despite urologic appearances.

There is complete absence or incomplete formation of the rectus abdominis and other muscles, which leads to the wrinkled abdominal wall of the prune infant (see Fig. 52.10C). This gives way to a fairly smooth “pot belly” in later life (Fig. 52.9). Reconstructive surgery is not normally required. The patients grow up physically active and strong but cannot sit up directly from a supine position. Abnormalities of the thoracic cage, such as pectus excavatum, are common.

Although true outflow obstruction may be present, the gross and irregular dilation of the urinary tract characteristic of PBS is primarily caused by a developmental defect with a variable degree of smooth muscle aplasia leading to aperistaltic ureters (Fig. 52.10A–B).

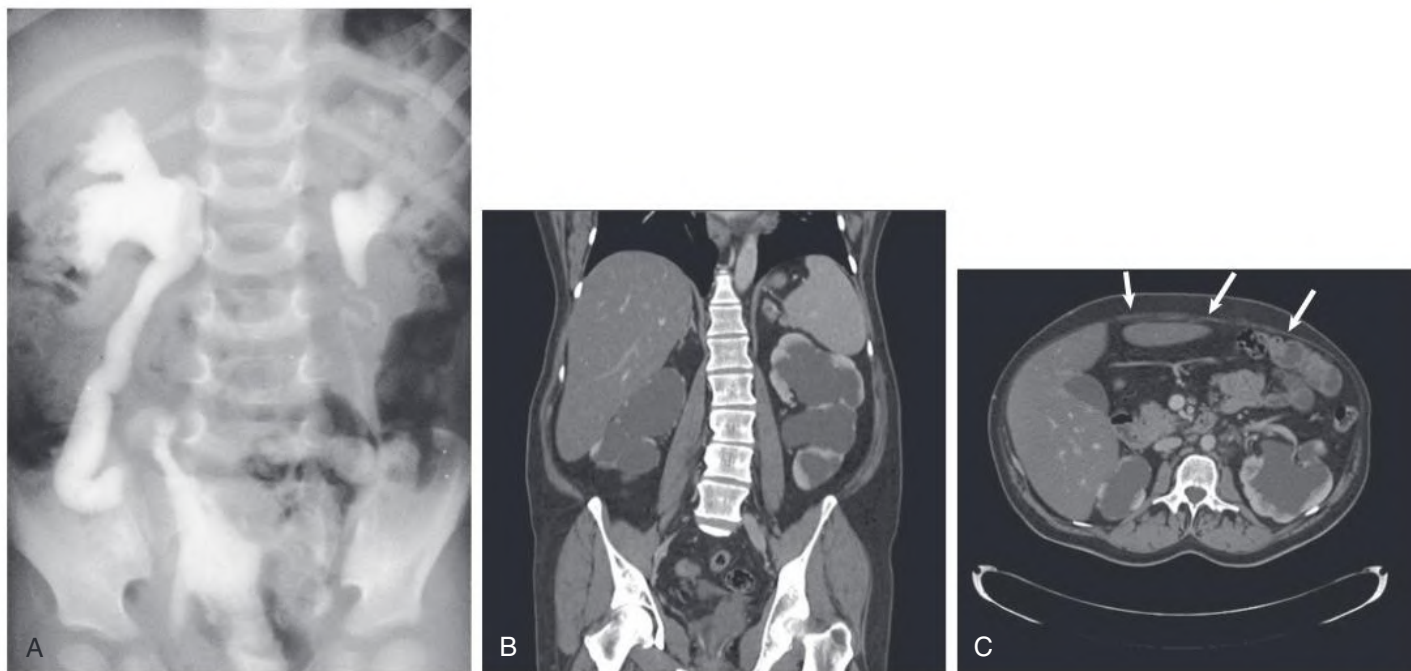


Fig. 52.10 Prune-Belly Syndrome. (A) Typical intravenous urogram appearance of a patient with prune-belly syndrome and good kidney function. Often, the ureters are extremely dilated and tortuous. (B) Computed tomography urogram showing gross bilateral hydronephrosis. (C) Absence of anterior abdominal wall musculature (arrows).

Urodynamic studies are often difficult to interpret because of gross VUR, but typically there is a low-pressure bladder with significant residual urine volumes. With late manifestation, some patients have detrusor instability. UTIs are common as a result of VUR and urinary stasis with constipation resulting from a defective Valsalva maneuver also increasing susceptibility.

Differential Diagnosis

In severe cases of megacystis or megaureter with gross impairment of kidney function (often with dysplastic kidneys), the differential diagnosis of PBS includes PUV, kidney dysplasia with or without multiple congenital defects, or neuropathic bladder.

Natural History

Once any outflow obstruction is addressed, usually in infancy, the GFR should remain stable despite the frightening radiologic appearances. In PBS patients observed in our unit for up to 40 years, kidney deterioration and hypertension have been rare. In the small number who have progressed, recurrent infection, hypertension, and proteinuria have been warning signs of impending trouble. Kidney scarring should be assessed by isotopic DMSA scintigraphy and kidney function followed by serial isotopic GFR measurements. Lifelong attention to blood pressure, UTIs, and stones is necessary.

Treatment

In all children with PBS, even with good kidney function, a careful search for obstruction should begin with the urethra and work up to the PUJ. Often, however, no obstruction is found, and no surgery is required. In many other patients, the floppy bladder is not anatomically obstructed, but bladder emptying is improved by urethrotomy ("functional obstruction"). In infancy, there is debate about the need

for reconstructive surgery. Certain patients born with severely compromised kidney function require reconstruction after stabilization by early diversion.

The current view is that the testes should be brought down to the scrotum (orchiopexy) in infancy in the hope that earlier surgery will produce proper germ cell development and thus preserve fertility. Although men with PBS have azoospermia, and are by definition infertile, there are case reports of successful paternity using intracytoplasmic sperm injection.

Bladder Exstrophy (Ectopia Vesicae)

Classic bladder exstrophy is the failure of the anterior abdominal wall and bladder to close, resulting in an open, inside-out appearance. These anomalies form part of the bladder exstrophy-epispadias complex (BEEC), which range from epispadias of an otherwise normal penis to major cloacal abnormalities (Fig. 52.11). The condition occurs in 1 in 50,000 births. The male-to-female ratio is 2:1.

Pathogenesis

Failure of growth of the lower abdominal wall between the allantois and the urogenital membrane coupled with breakdown of the urogenital membrane leaves a small, open bladder plate, a low-placed umbilical root, and diastasis of the pubic bones (Fig. 52.12B). The genital tubercle is probably placed lower in these patients, and the cloacal membrane ruptures above it, leading to a penis with an open dorsal surface that is continuous with the bladder plate. A midline closure defect causes a failure of fusion of the lower anterior abdominal wall, including the symphysis pubis, lower urinary tract, and external genitalia. Familial clustering suggests a genetic basis for the disorder, and association with the *ISL1* locus and 22q11.2 duplications have been reported.²⁷

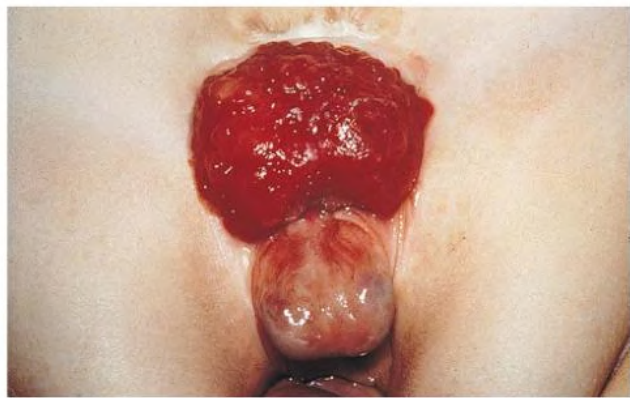


Fig. 52.11 Bladder Exstrophy. The entire length of the penis is also open (epispadias). (Courtesy Prof. C.R.J. Woodhouse, University College Hospital, London.)

Clinical Manifestations

In severe cases, the bladder mucosa lies exposed on the lower abdominal wall, with the bladder neck and urethra laid open. The prostate and testes are normal. Most patients have normal kidneys at birth, but approximately 18% are reported to have duplex, ectopic, or hypodysplastic kidneys.²⁸ Kidney function may be preserved after urinary diversion, although reflux is common (see Fig. 52.12A). Other congenital abnormalities affecting the heart, abdomen, skeleton, and joints are seen in approximately 50%.²⁸ More severe cloacal abnormalities are associated with imperforate anus and high or low rectal atresia.

Natural History

Long-term kidney outcome depends on the bladder. In a study with 12-year follow-up, the kidneys survive much better with a well-functioning bladder; 13% of those with a good bladder had significant kidney damage compared with 82% with ileal conduits, 22% with non-refluxing colonic conduits, and 33% with ureterosigmoidostomy.²⁹ Currently, the bladder is most commonly augmented (enterocystoplasty, ileocystoplasty, caecocystoplasty) or occasionally replaced by bowel (intestinal reservoir). In a study of 53 such patients monitored for more than 10 years, kidney function deteriorated (GFR decrease $\geq 20\%$) in only 10 patients.³⁰

Treatment

When the infant is born, the three urologic treatment goals are to close the abdominal wall, establish urinary continence to preserve kidney function, and reconstruct cosmetically acceptable genitalia. Surgery can be performed using either a staged or complete primary repair approach. The aim of initial surgery is to close the bladder, posterior urethra, and abdominal wall and convert the defect to a complete epispadias (Fig. 52.13). Repair of the epispadias is carried out later in infancy followed by reconstruction of the bladder neck and bilateral ureteral implantation at the age of 4. If the bladder is small, intestinal augmentation is required. Patients may be able to void, but many have to use catheters. Incontinence may be a long-term problem.

Neurogenic Bladder

In childhood, the most common cause of a neurogenic bladder is myelomeningocele, although with antenatal diagnosis and termination of affected pregnancies, as well as folate supplementation, it is becoming less common. A neurogenic bladder may also be seen without associated neurologic or other obvious causes (Table 52.3). The principal

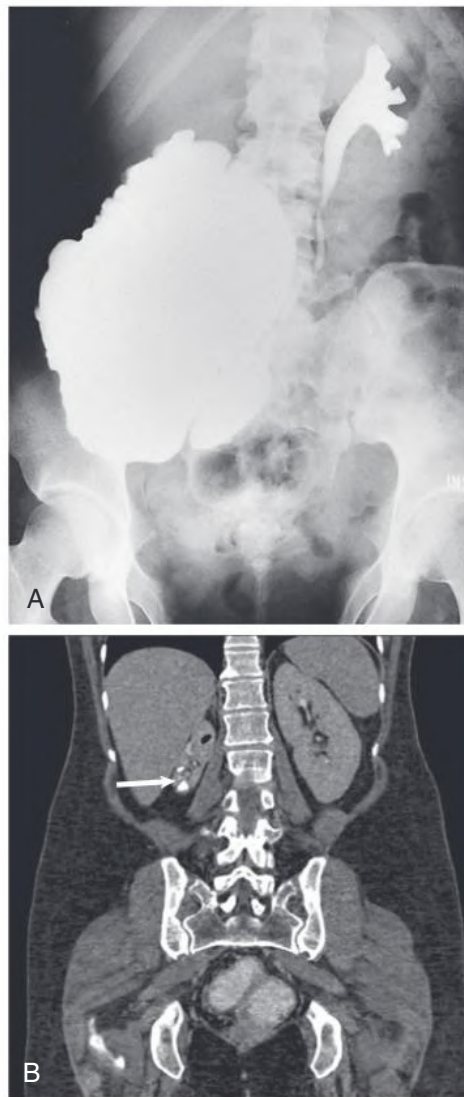


Fig. 52.12 Cystography in Bladder Exstrophy. (A) A 26-year-old woman with bladder exstrophy who has a continent Mitrofanoff system (see Fig. 52.18) with use of the colon to create a reservoir. There is reflux into the left kidney. Reflux also occurs into the right kidney, but the kidney is obscured by the full reservoir. Glomerular filtration rate is 130 mL/min. (B) Magnetic resonance scan of a patient with bladder exstrophy shows a scarred, small left kidney with several kidney calculi (arrow). Note widely splayed symphysis pubis.

consequences are incontinence, infection, and reflux with upper tract dilation and subsequent kidney failure. Early urodynamic assessment is essential (Fig. 52.14). Three different patterns of bladder behavior are seen: contractile, intermediate, and acontractile.

Contractile Behavior

An overactive detrusor (hyperreflexia) can produce some bladder emptying (incontinence). Unfortunately, 95% of patients have sphincter dyssynergia (inability to relax the urethral sphincter), which results in no relaxation and incomplete emptying of the bladder. Patients with incomplete lesions may have some control of the distal sphincter and normal anal and sacral reflexes. Ironically, although this latter group has the least neurologic deficit, they have the worst bladder situation, generating high pressures and great risk for kidney injury. The bladder becomes progressively hypertrophic, fibrotic, and poorly compliant.



Fig. 52.13 Epispadias. Result of multiple surgeries to close the epispadias and to lengthen the penis. (Courtesy Prof. C.R.J. Woodhouse, University College Hospital, London.)

TABLE 52.3 Causes of Neurogenic Bladder

Site of Lesion	Causes
Cerebral	Cerebrovascular accident, cerebral palsy, encephalopathy, trauma, Parkinson disease, dementia
Spinal	Isolated (no other neurologic features), trauma, multiple sclerosis, compression, spina bifida, spinal dysraphism, tethered cord, sacral agenesis, sacral teratoma
Peripheral nerve	Pelvic surgery, diabetes

Botulinum toxin A can be injected into the bladder wall to improve compliance and reduce bladder pressure; it is increasingly being used in children with neurogenic bladders to delay or avoid the need for augmentation cystoplasty.

Intermediate Behavior

These patients have some detrusor activity but not sufficient to empty the bladder. These intermediate bladders are poorly compliant, and patients have no voluntary control of their sphincters. Any rise in bladder pressure tends to cause incontinence, or the high intravesical pressures lead to kidney injury.

Acontractile Behavior

About 25% of patients have no detrusor activity, and the bladder overflows when it is sufficiently full. This acontractile bladder is not usually associated with kidney failure.

Myelodysplasia

Myelodysplasia refers to a group of neural tube anomalies that primarily affect the lumbar and sacral segment of the spinal cord and are the most common cause of neurogenic bladder dysfunction in children. *Spina bifida* reflects defective fusion of the posterior vertebral arches, of which there are several different types. *Meningocele* implies that the meninges extend beyond the confines of the vertebral canal with no neural elements contained inside. The most severe type of spina bifida is *myelomeningocele*, which has neural tissue protruding with the meningocele. Closed *spinal dysraphism* (spina bifida occulta) defines a group of structural anomalies of the caudal end of

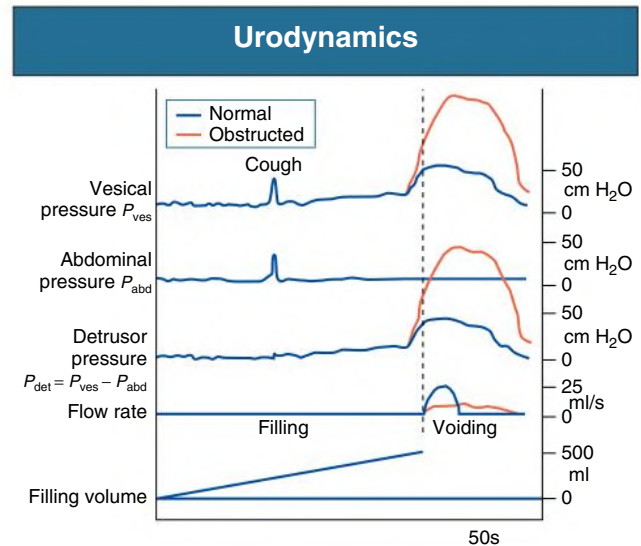


Fig. 52.14 Urodynamic Assessment by Cystometry. The vesical pressure is measured simultaneously with the abdominal pressure through the rectum; the detrusor pressure is the difference. A cough is used as a marker to show that the system is working. During filling, the first desire to void is normally at a detrusor pressure of <10 cm H₂O. This point is noted. The voiding pressure should normally be <40 cm H₂O (and is lower in women). Detrusor instability is an unstable (spontaneous) contraction occurring with a detrusor pressure >15 cm H₂O. Higher pressure can cause incontinence. In combination with radiologic imaging (videocystometry), the following are recorded: bladder neck, closed or open; bladder pressure, end filling; voiding detrusor pressure; bladder stability; compliance; flow rate, maximum; sensation, first; volume, voided and residual. (Courtesy Prof. M. Craggs, University College London.)

the spinal cord that do not result in an open vertebral canal but are associated with incomplete fusion of the posterior vertebral arches and are usually asymptomatic. *Sacral agenesis* is a rare anomaly in which part or all of two or more vertebral bodies is absent. It occurs early in fetal development when there is failure of ossification of the lowest vertebral segments, and is strongly associated with maternal type 1 diabetes.³¹ Partial sacral agenesis can be associated with an anterior meningocele.

Pathogenesis

The neural tube normally forms as the neural folds close over and fuse, starting in the cervical region and progressing caudally. The embryologic defect is likely an incomplete tubularization of the neural tube, with inadequate mesodermal invagination and subsequent arrest of vertebral arch formation.

The incidence of myelodysplasia varies from 1 to 7 in 1000 live births, but there are wide geographic variations. Monozygotic twins are often discordant for spina bifida, but siblings are at increased risk (1:10–1:20), and children of affected parents have a 4% chance of having a similarly affected child. Myelomeningocele accounts for more than 90% of myelodysplastic infants. Folic acid supplements taken during the first trimester reduce the incidence of myelodysplasia by 70%.³²

Clinical Manifestations

All causes of tethered cord can produce a variable neurologic deficit. During development, some children experience progressive neurologic disturbance with bladder dysfunction, bowel dysfunction, scoliosis, and a syndrome of pes cavus and limb growth failure. Learning

difficulties may also be seen in individuals with myelomeningocele, especially those who have required ventriculoperitoneal shunting for associated hydrocephalus.

Bladder dysfunction. Nearly all individuals with myelomeningocele have a neurogenic bladder, which may result in urinary incontinence and an adverse impact on the upper tracts. It can be an isolated problem with abnormal urodynamic studies but a normal neurologic examination.

Bowel dysfunction. Bowel dysfunction is often present and needs to be treated accordingly. There may be severe constipation and overflow incontinence. First-line management involves the use of laxatives, suppositories, and enemas, followed by transanal irrigation or antegrade continent enterostomy for those who have refractory symptoms. The appendix is brought out to the abdominal surface, and thus the colon can be irrigated antegrade with saline.

Natural History

About 14% of patients have kidney complications at birth and are at high risk in the next few years. Ultimately, 26% will develop CKD and 1.3% ESKD by early adulthood, with continence achieved in 37%.³³ Poor kidney outcome can be predicted by urodynamic findings, with worst outcomes related to increased bladder wall thickness, degree of reflux, urethral pressures above 70 cm H₂O, and reduced bladder capacity (Fig. 52.15). VUR occurs in 3% to 5% of newborns with detrusor hypertonicity or dyssynergia. Without treatment, this increases to 30% to 40% by age 5 years.³⁴

Treatment

Management of the bladder depends on the urodynamic findings. In the 1970s, clean intermittent self-catheterization was introduced, but before that time, urinary diversion was the usual treatment. Currently, management is early initiation of intermittent self-catheterization, and antimuscarinic drugs (e.g., oxybutynin) are used for high pressure, hyperreflexic bladders with VUR to increase bladder compliance. Intravesical administration of oxybutynin should be considered in individuals with severe side effects (dry mouth, facial flushing, blurred vision, heat intolerance). With persisting symptoms related to bladder storage or detrusor overactivity, bladder wall injection with botulinum toxin type A1 is offered, with effects lasting 15 months on average.³⁵ In the presence of deteriorating kidney function or intractable symptoms, ileal or colonic bladder augmentation and creation of continent catheterizable channels may be required.³⁶

Bladder Outflow Obstruction

Congenital bladder outflow obstruction is rare and is usually caused by a neurogenic bladder, PUV, or an ectopic ureterocele.

Posterior Urethral Valves

PUV are the most common cause of severe subvesical obstruction, occurring in 1 in 4000 male births.³⁷ As a result, bilateral hydronephrosis and megaureter occur. Obstruction is caused by a diaphragm that extends from the floor to the roof of the urethra at the apex of the prostate. Valves appear as mucosal folds in the posterior urethra below the verumontanum. There is dilation of the proximal urethra and bladder wall hypertrophy and trabeculation. Above the valves, the prostatic urethra dilates, undermining the bladder neck. The valves obstruct flow only in one direction, and therefore a catheter can be passed without difficulty.

Pathogenesis

The urethra develops in two parts: differentiation of the urogenital sinus (posterior urethra) and tubularization of the urethral plate

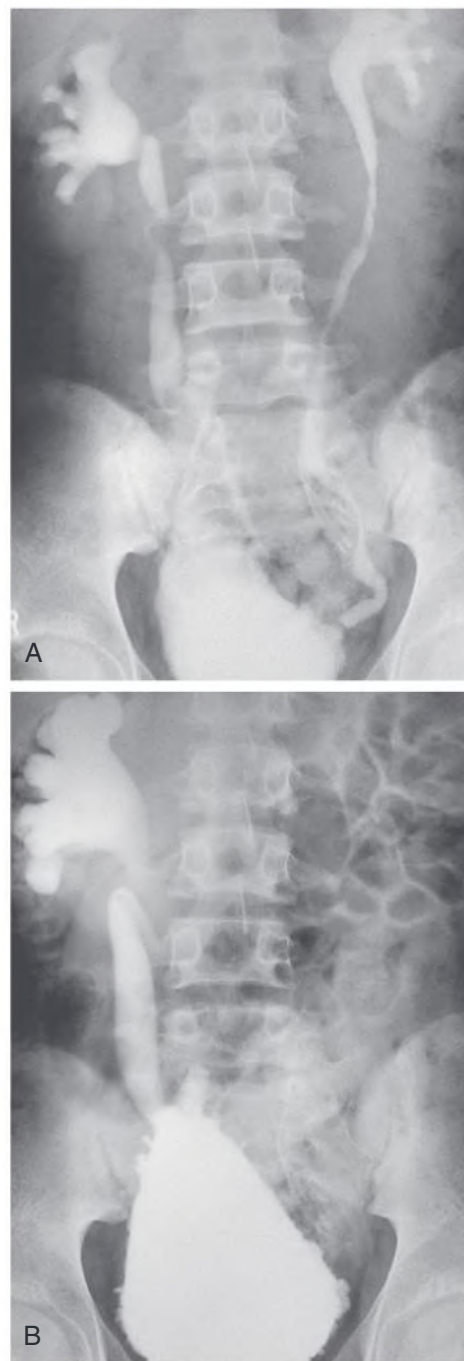


Fig. 52.15 Sacral Spina Bifida With Neuropathic Bladder. (A) Intravenous urogram shows evidence of a previous hydronephrosis and subsequent scarring of the right kidney. The architecture of the left kidney is well preserved. (B) Micturating cystogram. The typical tapering, hypertrophied, trabeculated bladder gives the characteristic “pinecone” appearance. Note the gross reflux on the right side. This is probably helping to protect the left kidney by acting as a “pop-off” mechanism. This is analogous to the protection that can occur in males with posterior urethral valves.

(anterior urethra). Persistence of the urogenital membrane and abnormal integration of the mesonephric (Wolffian) duct into the posterior urethra have been proposed as potential mechanisms underlying PUV. Monogenic defects appear to be rare, although heterozygous variants in *BNC2*, a zinc finger protein expressed in the embryonic urethra, have been identified in familial PUV.²⁷

Clinical Manifestations

Most cases of PUV are detected antenatally on US as evidenced by bilateral hydronephrosis, dilated bladder, posterior urethra (keyhole sign), and, in severe cases, oligohydramnios. Infants present with a palpable distended bladder and enlarged kidneys, abnormal urine stream, or failure to thrive as a result of kidney failure. Kidney dysplasia is common and around a third of children also have VUR.³⁸ Children with less severe disease present with poor stream, incontinence, UTI, or kidney failure. However, late presentation is also associated with worse outcome.³⁹

Three abnormal features can help protect the kidney, reducing the high pressures generated during voiding: massive unilateral reflux, usually with ipsilateral kidney dysplasia (thereby protecting the other kidney); large bladder diverticulum; and urinary extravasation, often with urinary ascites. These protective mechanisms are referred to as *pop-off mechanisms* (see Fig. 52.15B).⁴⁰ US can show the bladder thickening, dilated system, and dilation of the posterior urethra. A specific diagnosis should be documented by videocystometrography (see Urodynamics).

Natural History

In the 1960s, 25% of children with PUV died within the first 12 months, and 25% died later in childhood, including fatalities because of ESKD. By the late 1990s, early mortality was less than 5%,⁴¹ and life-time risk of ESKD is now estimated at 28.5%, reached in the majority before the age of 30.⁴² The bladder may become stretched, resulting in poor emptying, or unstable, leading to poor compliance, unsuppressed detrusor contractions, and high storage pressure. Both situations are exaggerated by progressive polyuria. Such patients may have a daily urine volume of 5 L. Urodynamic follow-up studies suggest that instability decreases with time; bladder capacity increases, but there are unsustained voiding contractions. Prognosis correlates with the nadir serum creatinine value once obstruction has been relieved and with the degree of bladder dysfunction.⁴³ Despite adequate early treatment, CKD as a result of bladder dysfunction and associated kidney dysplasia develops in many children.^{42,43}

Treatment

All children have endoscopic transurethral resection of their valves in infancy. Bladder diversion should be avoided. Bladder instability and poor bladder compliance must be treated, regardless of whether symptoms result. Males with substantial residual volumes can be managed by clean intermittent self-catheterization, but adherence is often poor because of urethral discomfort or because previous urethral surgery has made the passage of catheters difficult. Adherence is a particular problem with adolescents who are continent and for whom kidney failure is too abstract a concept. Continence often improves spontaneously at puberty but can be helped by imipramine. Deterioration in GFR will require further examination of urine flow rate and exclusion of urethral stricture.

Urethral Diverticulum

Urethral diverticulum usually occurs in males and is rare. It may manifest with UTI, obstruction, or stones. The two types are anterior and posterior. The anterior type can be associated with anterior urethral valves and obstruction.

Other Congenital Causes of Bladder Outflow Obstruction

Urofacial Syndrome

Urofacial syndrome (UFS), or Ochoa syndrome, is a rare autosomal recessive disease characterized by facial grimacing when attempting to smile and failure of the urinary bladder to void completely, despite a lack of anatomic bladder outflow obstruction or overt neurologic

BOX 52.1 General Principles of Management of Congenital Urinary Tract Abnormalities

- Educate to encourage adherence.
- Review urologic status.
- Find cause of urinary tract obstruction and treat.
- Control blood pressure.
- Monitor kidney function and proteinuria.
- Treat acidosis.
- Prevent bone disease.
- Check for stones.
- Institute clean intermittent self-catheterization for chronic urinary retention.
- Maintain bladder storage pressure <40 cm H₂O.
- Maintain bladder volume <400 mL.

damage.²⁷ Patients present with enuresis and UTI and all the features of a neurogenic bladder together with dilated upper tracts. They are at risk for kidney failure. Biallelic mutations of *HPSE2* and *LRIG2* have been reported. The proteins they encode are detected in nerves found in the fetal bladder, implying UFS may be a peripheral neuropathy of the bladder.²⁷ UFS is part of a spectrum of congenital bladder disorders that include nonneurogenic neurogenic bladder or Hinman syndrome.

Gonadal Dysgenesis

Disorders of sex development, gender identity, and micropenis are beyond the scope of this chapter and rarely encountered in adult practice. Patients will be met, however, with ESKD who are phenotypically female but are genotype XY and have mutations of *WT1* (Denys-Drash and Frasier syndromes). They have gonadal dysgenesis and must have their streak ovaries removed; otherwise, gonadoblastomas will develop.

GENERAL MANAGEMENT OF CONGENITAL KIDNEY AND URINARY TRACT ABNORMALITIES

The principles of management of congenital urinary tract abnormalities are shown in Box 52.1. The most important part of the management is ensuring that the patient, family, and primary care physician know what can and must be done. First, they must understand the necessity of long-term follow-up at least annually. ESKD often occurs when a patient is lost to follow-up, often manifesting later with accelerated hypertension and rapid loss of GFR.

Clinical Evaluation

By the time the adolescent attends an adult clinic, it is assumed that the urinary tract is not obstructed and that further surgery is not required. Nevertheless, it is the responsibility of the nephrologists and urologists who care for these young people to review this aspect periodically.

Symptomatic UTI is common and must be treated promptly. Increase in frequency or severity of infections must lead to investigations to find the cause.⁴⁴ The BP must be monitored regularly and kept normal. Finally, kidney function must be monitored, proteinuria assessed, and the cause of any deterioration identified. As in any kidney condition, the residual GFR may decline inexorably, which is associated with increasing proteinuria and hypertension. As with other kidney conditions, kidney function is usually stable when proteinuria is minimal or absent. Deterioration in the absence of proteinuria must alert the physician to the likelihood of obstruction or the adverse effect of a nephrotoxic drug.

TABLE 52.4 Monitoring Patients With Congenital Urinary Tract Abnormalities: Routine Investigations for Assessment of Clinical Status

Baseline Measurements	Reason for Test
Radiology	
CT KUB	Exclude stones
Ultrasound	
Ultrasound of kidneys	Assess kidney size and hydronephrosis
Ultrasound of bladder after micturition	Assess residual volume
Urine flow rate	Ensure adequacy
Scintigraphy	
Glomerular filtration rate: ^{51}Cr -labeled EDTA	Baseline kidney function
Dynamic isotope scan: $^{99\text{m}}\text{Tc}$ -labeled MAG3 or DTPA	Assess outflow obstruction
Static isotope scan: $^{99\text{m}}\text{Tc}$ -labeled DMSA	Assess scarring and divided function
Biochemistry	
Urine protein-creatinine ratio	Assess proteinuria

^{51}Cr , Chromium-51; CT, computed tomography; DMSA, dimercaptosuccinic acid; DTPA, diethylenetriaminepentaacetic acid; EDTA, ethylenediaminetetraacetic acid; KUB, kidney, ureter, bladder; MAG3, mercaptoacetyl triglycine; $^{99\text{m}}\text{Tc}$, technetium-99m.

The routine investigations performed to document the current situation act as a reference point for the future (Table 52.4). If the bladder empties completely with an adequate flow rate (15 mL/s), no problems should arise. If there is any doubt about the condition of the bladder, urodynamic investigations are necessary. If the clinical situation changes, further investigations are required. An increase in UTIs might suggest a stone or increase in residual urine. With an unexpected decline in kidney function, obstruction again must be excluded.

The patient should keep a 24-hour urine volume diary every 6 to 12 months, recording the time of voiding and volume passed. It is best to ask patients to do this on 2 consecutive days to determine the maximum bladder capacity and the total 24-hour urine volume. This should be done before urodynamic investigations because results can be misleading if the bladder is not filled to capacity.

Exclude Obstruction

Obstruction must always be excluded if there is a decline in GFR. The possibility of obstruction may be raised by a routine US (see Fig. 52.7) and should be pursued with $^{99\text{m}}\text{Tc}$ -MAG3 scintigraphy to exclude obstruction (Fig. 52.16).

In patients with conduits, obstruction can be excluded by infusion of contrast material into the loop (loopogram) and demonstration of reflux up the ureter.

Rarely, in patients with large bladders or in transplant recipients, the kidney may become obstructed when the bladder reaches a certain volume. This can be investigated by filling the bladder by catheter and performing $^{99\text{m}}\text{Tc}$ -labeled MAG3 scintigraphy, initially with the bladder full. If there is no excretion, the bladder volume can be reduced in 100-mL increments until there is flow down the ureter (Fig. 52.17).

Urodynamics

Any urodynamic investigation should start with a free urine flow rate. Provided the flow rate is normal and the bladder empties completely

(leaving no residual volume on postmicturition US), it can be assumed that there is no significant bladder outflow obstruction.

Complete investigation of abnormalities of bladder and urethral function requires synchronous recordings of intravesical and intrarectal pressures taken during bladder filling and emptying (see Fig. 52.14). Combined with radiologic imaging, the study is known as *videocystometrography* (VCMG).

Surgical Correction of the Urinary Tract

A normal bladder acts as a low-pressure, good-volume urine reservoir that is continent, is sterile, and empties freely and completely. Any other form of urine reservoir aims to recreate such an environment. When this is not achieved in either a natural or a reconstructed bladder, complications such as sepsis and kidney dysfunction can occur.

A variety of conduits and continent reservoirs have been developed to replace unusable bladders. Ileal conduit diversion has been most widely used for native kidneys, although deterioration in kidney function frequently results from long-term complications, including urosepsis, kidney calculi, and, most often, stenosis, leading to obstruction or reflux with ureteral dilation. There is an overall complication rate of 45%, but with a high index of suspicion and an aggressive diagnostic and therapeutic approach, many of these problems can be detected and treated early, with resultant good long-term function of native kidneys. Similar results may be obtained when kidney transplantation is performed in these patients.⁴⁵ Other forms of urinary diversion that are continent and therefore more socially acceptable to patients are now widely used in general urologic practice and are being encountered in kidney transplantation (see Fig. 52.12A). These forms include augmented bladders draining through the urethra and augmented or intestinal bladders draining through continent stomas.

COMPLICATIONS

Urinary Tract Infections

Symptomatic UTIs are common. Risk factors include stagnation of urine, stones, foreign bodies (stents, catheters), previous infections, and kidney scarring. UTIs must be treated promptly after a urine culture specimen (midstream or catheter specimen) has been taken. Recurrent UTIs, particularly after a period of stability, must lead to further investigations to exclude stones or obstruction, including CT kidney, ureter, bladder (KUB); kidney US, and postmicturition bladder US.

Asymptomatic UTIs often do not require treatment (except during pregnancy). For patients with urinary diversions, it is important to obtain a catheter specimen of urine because urine taken from a bag is invariably infected. Cloudy, malodorous urine should be treated initially with increased fluid intake and frequency of self-catheterization (every 1–2 hours for 12 hours).

It is sometimes appropriate to give prophylactic antibiotics, such as trimethoprim or nitrofurantoin, in the context of recurrent UTIs ($\geq 3/\text{yr}$). Nitrofurantoin should be avoided if eGFR is less than 45 mL/min/1.73 m² because its efficacy relies on excretion into the urinary tract. Quinolones should not be used for prophylaxis, if possible, because of the risk for inducing resistance. Antibiotic prophylaxis in children has been shown to reduce the incidence of repeat symptomatic UTI but has little impact on “kidney scarring.”⁴⁶ When foreign bodies such as stones remain, attempts to sterilize the urinary tract are unlikely to be successful. If prophylactic antibiotics are no longer effective at preventing infection, it is advisable to stop all antibiotics and give the patient a supply of antibiotics to treat symptoms arising at home.

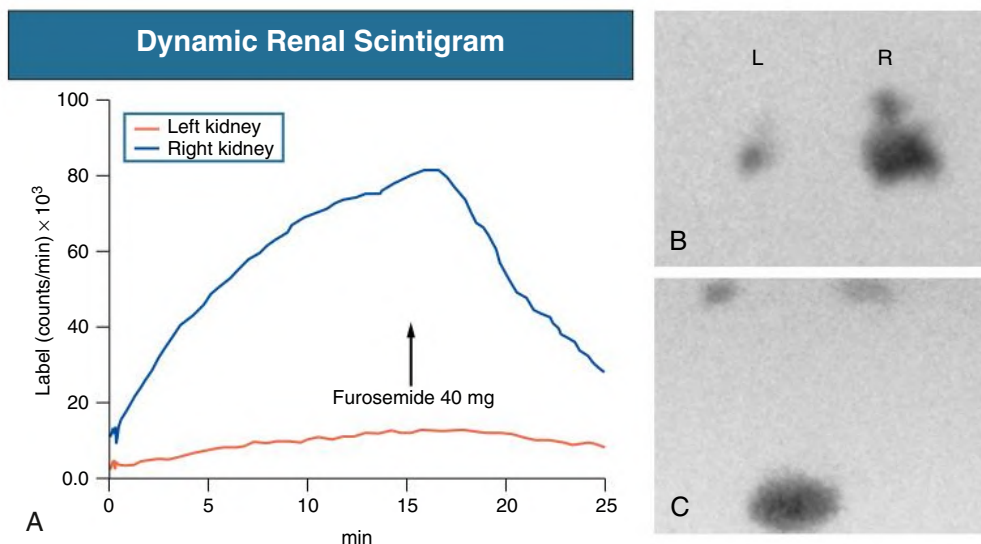


Fig. 52.16 Dynamic ^{99m}Tc -Labeled Mercaptoacetyltryglycine Kidney Scintigram. (A) Time-activity curve showing accumulation of isotope in right kidney that washes out after furosemide, thus excluding significant obstruction. (B–C) Images from the same study showing holdup of isotope in dilated right (R) renal pelvis (B) that washes out into the bladder after furosemide (C), excluding significant obstruction.

Single Kidney Obstruction by Full Bladder

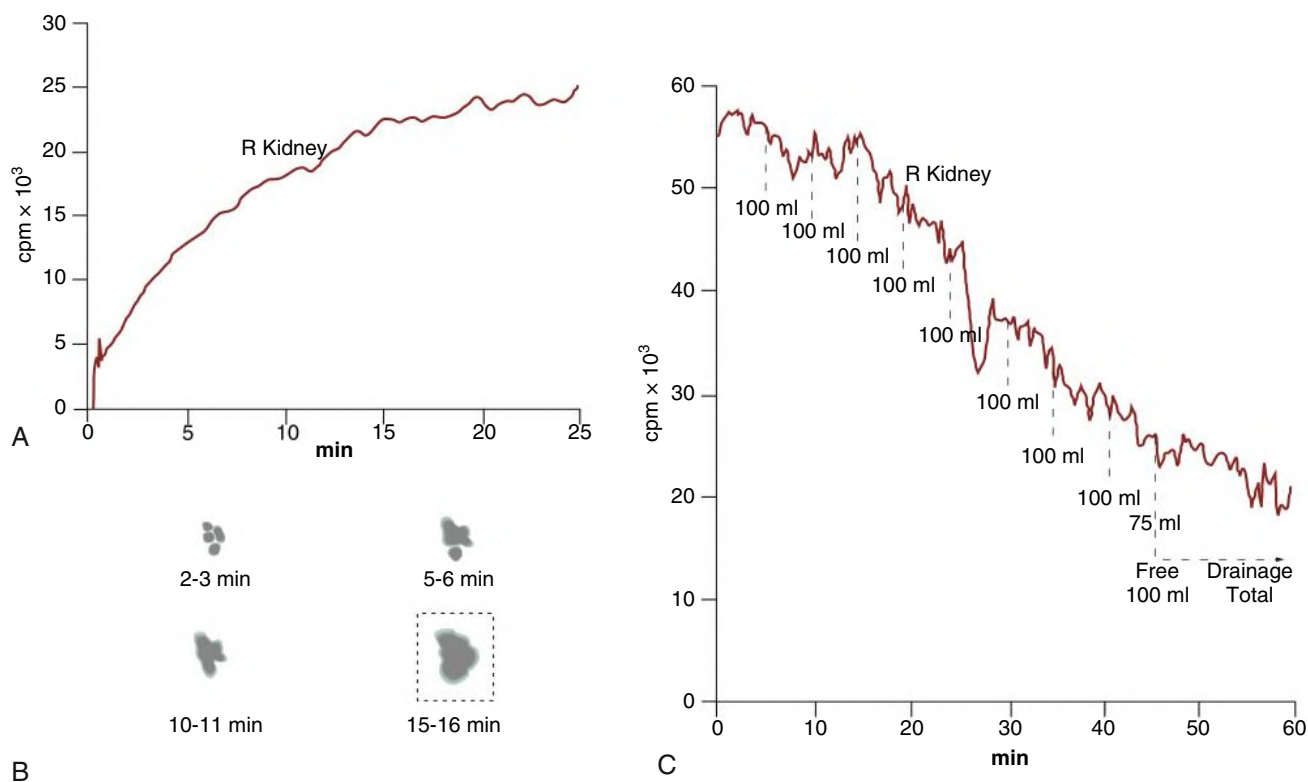


Fig. 52.17 Dynamic Isotope Scan (Mercaptoacetyltryglycine) Starting with Bladder Full in a Patient With a Solitary Right Kidney. (A) Rising curve of tracer accumulating in kidney and showing no excretion. (B) Accumulation of isotope in hydronephrotic pelvis without excretion to bladder. (C) The 100-mL increments of fluid removed from bladder result in eventual free drainage of the kidney. cpm, Counts per minute.

Glomerular Hyperfiltration

If kidney function is declining with proteinuria and hypertension, glomerular hyperfiltration is likely, although all other causes of kidney dysfunction must be excluded. Patients should be treated with RAS blockade with angiotensin-converting enzyme (ACE) inhibitors or angiotensin receptor blockers (ARBs).

Proteinuria and Progressive Kidney Failure

Can progression to ESKD be predicted, and does treatment with ACE inhibitors delay or prevent this? We investigated this in a retrospective review of patients with scarred irregular kidneys caused by primary kidney dysplasia or abnormal bladder function.^{47,48} All patients had at least 5 years of follow-up, and when ACE inhibitors were started, eGFR was less than 60 mL/min/1.73 m² (mean, 41 mL/min), with mean proteinuria of 1.7 g/24 h. ESKD developed in 46% of patients but in none with proteinuria less than 0.5 g/24 h and in only 2 of 18 patients with eGFR greater than 50 mL/min/1.73 m².⁴⁷ The outcome of the two groups was similar whether there was primary kidney dysplasia or abnormal bladder function. ACE inhibitor treatment slowed decline in kidney function at all levels of function.⁴⁸ The similar outcome of the two groups indicates that progressive kidney failure in young males born with abnormal bladders is caused by intrinsic kidney pathophysiologic processes, in contrast to the view that it is a result of poor bladder function.

Hypertension

Hypertension is common in the presence of scarred kidneys, but it is usually controlled easily with one or two drugs. Patients in whom CKD is secondary to obstruction tend to have volume contraction and therefore often have normal BP or only mild hypertension. ACE inhibitors or ARBs are preferred for patients with proteinuria and progressive kidney failure. Diuretics should not be used if the patient is volume-contracted.

Stones

Stones that form in the presence of infected urine are typically magnesium ammonium phosphate (struvite) or calcium phosphate (hydroxyl apatite, carbonate apatite, calcium hydrogen phosphate [brushite], tricalcium phosphate [whitlockite]). These salts are poorly soluble in alkaline urine. In 90% of patients, the infecting organism is *Proteus* spp., but other urea-splitting organisms (including some staphylococci and *Pseudomonas* spp.) also generate ammonia.

Stones, usually calcium phosphate, are common in conduits because of the alkaline environment and occur in 5% to 30% of ileal conduits.⁴⁹ Stones must be suspected if UTIs recur or become more frequent, if kidney function suddenly deteriorates, or if there is an unexplained sterile pyuria.

Tubular Dysfunction

Patients whose kidney failure is secondary to obstruction have significant tubular injury. This may cause problems, in particular with urinary concentration, acidification, and sodium reabsorption.

Polyuria

Nocturia is one of the most significant symptoms in the assessment of patients in whom obstruction or tubular dysfunction is suspected. Overfilling of the bladder or reservoir is an important cause of intermittent upper tract obstruction and deteriorating function. The 24-hour urine volume diary is a simple way to assess this.

Salt Depletion

Patients with tubular damage may have a salt-losing tendency. Patients typically have a cool periphery and constricted hand veins with no peripheral edema. Increasing salt intake can relieve cramps, improve kidney function, and reduce hyperuricemia, but at the cost of increasing BP. With patients who are salt depleted, it is important to give sodium chloride because it is the chloride anion that is deficient and responsible for the reduction in circulating volume.

Metabolic Acidosis

There is often a metabolic acidosis disproportionate to the degree of kidney impairment, particularly in individuals with a ureterosigmoidostomy (see later). This is secondary both to a proximal tubular failure of bicarbonate reabsorption and a distal tubular failure to secrete hydrogen ions. It is our practice to give sufficient sodium bicarbonate to correct the serum bicarbonate into the normal range.

Bone Disease

In addition to the typical bone disease of progressive CKD, acidosis contributes significantly to osteomalacia. Growing children are particularly vulnerable to osteomalacia, and great care must be taken to correct acidosis and actively manage bone disease.

Urinary Diversions

Ureterosigmoidostomy

Fortunately, it is now rare to meet a patient who has a ureterosigmoidostomy, which was widely used as a technique for urinary diversion until the 1970s. The ureters were anastomosed directly into the sigmoid colon with no disruption of bowel continuity. This technique was most often used in patients with bladder exstrophy. Although patients start with normal kidney function, there is frequently deterioration in function. In one series of 25 patients, significant kidney damage occurred in 50%. Stones, infection, and ureteral strictures are common, and patients remain at risk for colonic carcinoma, with a 10% incidence of carcinoma at 20-year follow-up.⁵⁰ However, this diversion is probably best known for the hyperchloremic, hypokalemic acidosis that occurs. Once the urine is in contact with the colonic mucosa, the urinary sodium exchanges for potassium and the chloride for bicarbonate via the SLC26A3 anion exchanger, with large quantities of ammonium ions produced by the action of fecal bacteria on urinary ammonia. Ammonium ions are absorbed both with chloride and in exchange for sodium. The severe acidosis is caused by ammonium ion retention and stool loss of bicarbonate. Patients are managed with large doses of oral sodium bicarbonate, which is titrated to keep plasma bicarbonate in the normal range (>22 mmol/L).

Ileal Conduits

Unlike the sigmoidostomy, in which urine enters a reservoir, the ileal conduit is free flowing, with rapid urinary transit and no reservoir. Therefore, metabolic complications are much less common, although again the bowel can exchange sodium and chloride for potassium and bicarbonate. B12 deficiency and diarrhea secondary to impaired bile acid absorption can occur. A number of other complications of ileal and colonic conduits can lead to progressive loss of kidney function (Box 52.2).

Enterocystoplasty and Intestinal Urinary Reservoirs

In 53 patients with bladder exstrophy and intestinal reservoirs, monitored more than 10 years with serial isotopic GFRs, kidney function deteriorated (GFR decrease $\geq 20\%$) in only 10 (~20%).³⁰ Loss of GFR

was caused principally by chronic urinary retention with or without infection in poorly compliant patients who did not catheterize regularly. Patients also must be checked regularly to ensure that anastomotic stenoses and high-pressure reservoirs do not occur. Stones are common and occur in up to 52% of patients.⁵¹ An increased risk of malignancy is also seen in augmentation cystoplasty, predominantly adenocarcinomas occurring at the entero-urinary anastomosis after a median of 19 years. Macroscopic hematuria should trigger prompt referral for cystoscopy.

BOX 52.2 Long-Term Complications of Urinary Diversion

- Pyelonephritis and recurrent urinary tract infections
- Calculi
- Obstruction
- Strictures
- Bladder mucus causing obstruction
- Cancer at intestinal-ureteral anastomosis
- Hyperchloremic acidosis
- Delayed linear growth in children
- Effects of intestinal loss from gastrointestinal tract (e.g., vitamin B12 deficiency)
- Complications related to abnormal pelvic anatomy (e.g., in pregnancy)
- Psychological and body image problems

END-STAGE KIDNEY DISEASE AND TRANSPLANTATION

This group of patients presents two important problems if they develop ESKD. First, because of multiple abdominal surgeries, continuous ambulatory peritoneal dialysis (CAPD) is often challenging, although if there is any doubt and the patient is interested, CAPD should be attempted. Second, the bladder and urinary reservoir must be suitable for kidney transplantation. If a bladder has just destroyed two good native kidneys, it is likely to do the same to a transplant kidney. Most patients will be maintained on hemodialysis, but it is frequently difficult to establish a good arteriovenous fistula because of chronic hypovolemia and venoconstriction. Patients receiving dialysis often continue to pass 1 L or more of urine per 24 hours, and they also remain at risk for serious UTI and pyelonephritis.

Pretransplantation Assessment

Transplantation into the abnormal lower urinary tract requires careful evaluation and follow-up. Thorough preoperative assessment of bladder function is essential (Fig. 52.18). Patients considered to have normal bladders require at least postmicturition bladder US and urinary flow rate.

All patients with abnormal bladders or reservoir must have a full VCMG to ensure that the bladder reservoir is large and adequately compliant. If the bladder is small or has not been used for some time, bladder cycling, which involves periodically filling and distending the

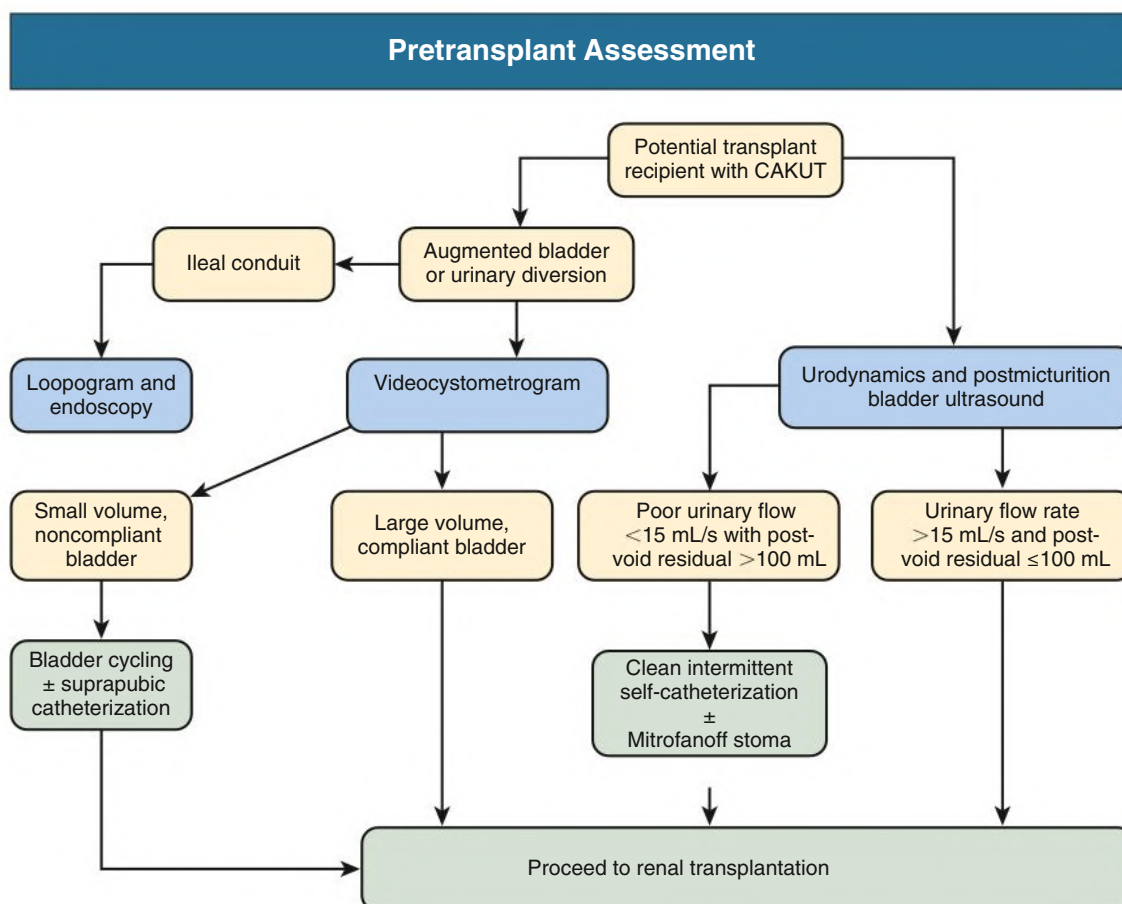


Fig. 52.18 Pretransplant assessment of patients with an abnormal lower urinary tract. CAKUT, Congenital anomalies of the kidney and urinary tract.

bladder through a suprapubic catheter, may be required. A urodynamic study before transplantation indicated that poor bladder function, as shown by small bladder volumes, is a predictor of graft loss even in patients with previously normal bladder function.⁵² Intermittent self-catheterization is safe and effective for a patient with a poor flow rate who fails to empty the bladder. This, however, is possible only with a normal urethra and a cooperative patient. When this is not practical, we attempt to establish suprapubic drainage through a continent channel, such as a Mitrofanoff stoma (Fig. 52.19). If a conduit is to be used, a loopogram and endoscopy must ensure it is in good condition. We do not remove native kidneys unless they are causing recurrent UTI.

Transplant Outcome

In an 18-year experience, we performed transplantation in 65 patients with abnormal bladders, with a total of 72 kidney transplants.⁵³ In 52 cases the ureters were transplanted into unaugmented bladders; in 20 cases, there was some form of augmentation or diversion. Results were compared with 59 transplants in 55 patients who had kidney failure from kidney dysplasia and whose bladder function was considered normal. There was no difference in actuarial graft survival in the two groups at 10 years (abnormal bladders, 66%; normal bladders, 61%), although longer follow-up showed an advantage for normal bladders, with graft survival of 29 to 33 years compared with 15 years for the abnormal bladders. UTIs were relatively common in all patients but produced problems only in patients with abnormal bladders. A recent study also reported comparable 10-year graft survival rates of 70% to 76% for individuals with kidney hypodysplasia and PUV, but an increased risk of graft loss for PUV after this time point.⁵⁴ This suggests that although short-term outcomes appear to be equivalent in individuals with normal and abnormal bladders, longer-term graft survival may be limited by bladder dysfunction.

Short-term outcomes after transplantation have been shown to be more favorable in both children and adults with CAKUT compared with individuals with primary glomerular or non-CAKUT diseases at 5 to 10 years, although recipients were more likely to be younger and receive a preemptive transplant.^{5,55,56} Young adulthood (15–24 years) is also a particularly challenging time, and the high rates of graft loss and eGFR decline observed during this period may be exacerbated by nonadherence to CISC in adolescents with bladder dysfunction.⁵⁷

Management

A double-J ureteral stent should be placed routinely at the time of transplant surgery. Adequacy of urinary drainage must be assessed frequently, even when graft function seems to be good. Two months after transplantation, when the ureteral stent has been removed, we recommend performing a baseline postmicturition US of the transplant



Fig. 52.19 Mitrofanoff Stoma. This patient was born with bladder exstrophy and has had a successful kidney transplant for 22 years. Her kidney is plumbed into a colonic reservoir, and she catheterizes herself through a continent Mitrofanoff stoma, which is covered by a small bandage in the photograph.

kidney and bladder, in addition to a dynamic ^{99m}Tc-MAG3 scintigraphy scan in patients with abnormal bladders.

US and ^{99m}Tc-MAG3 scintigraphy are then repeated when clinically indicated. The protein-to-creatinine ratio is measured on a random urine sample at every outpatient visit. If there is kidney dysfunction, imaging tests are repeated, and if there is a change from baseline, kidney biopsy is performed to exclude an immunologic cause of graft dysfunction. If there is deterioration in GFR in the absence of rejection or calcineurin inhibitor toxicity, the DMSA scan is repeated (to see whether new scarring has occurred) and the bladder reassessed urodynamically.⁵⁸

Complications

UTIs must be detected and treated early, and recurrent UTIs may require long courses of antibiotics or even removal of the native tracts. Symptomatic UTIs are common in the first 3 months after transplantation (63%); fever and systemic symptoms occur in 39% of patients with normal bladders and 59% with abnormal bladders. UTI directly contributes to graft loss in patients with abnormal bladders but causes no consequences in those with normal bladders.⁴⁴ Prophylactic administration of antibiotics for the first 6 months significantly reduces the subsequent incidence of UTI. When UTIs recur, a cause must be sought with ultrasound of the kidneys and bladder. A CT KUB can be done to look for stones in native or transplant kidneys and the bladder or urinary diversion. If there is a residual volume after double micturition, the patient must be instructed to perform clean intermittent self-catheterization. With these measures, good results are obtained.

SELF-ASSESSMENT QUESTIONS

- Which statement about unilateral kidney agenesis is *false*?
 - It occurs in 1 in 10,000 births.
 - Typically, there is no ureter.
 - The fallopian tube may be absent in females.
 - Kidney ultrasound is recommended in first-degree relatives.
 - It can be part of a syndrome.
- Which statement about prune-belly syndrome is *false*?
 - It usually occurs in males only.
 - The muscles of the anterior abdominal wall are absent.
 - Gross dilation of the bladder occurs, with normal ureters.
 - Males have bilateral undescended testes.
 - Patients are usually infertile.
- Ureterosigmoidostomy may lead to all of the following *except*:
 - Hyperchloremia
 - Hyperkalemia
 - Acidosis
 - Ammonium ion retention
 - Colonic carcinoma

-
4. Which of the following statements about posterior urethral valves is *false*?
- A. They are usually detected antenatally.
 - B. Patients require urethral surgery.
 - C. Progressive kidney failure is associated with proteinuria.
 - D. Posterior urethral valves may be confused with urofacial (Ochoa) syndrome.
 - E. Posterior urethral valves occur in males and females.
5. Which of the following statements about urinary tract infection (UTI) in an augmented bladder is *false*?
- A. The patient with UTI does not require treatment when asymptomatic.
 - B. UTI often causes calcium oxalate stones.
 - C. Recurrent UTIs should be investigated with ultrasound, including assessment of postvoid residual volume.
 - D. UTI may indicate poor bladder emptying.
-

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